DATA COMMUNICATION FINAL T1.2023

Q1 : Filling the approciate word(s) from the box into the following two charts :

(analog , sin waves , non periodic , periodic , composite , simple , analog , sine wave)

1) **Singal :** ***analog & sin waves & composite & non periodic***
2) **Singal :** ***analog & sine wave & simple & periodic***

Q2 : Choose the correct answer :

1) **Radio waves can penetrate walls, this property can considered as :**
   a) Advantage
   b) Disadvantage
   c) Advantage & Disadvantage
2) **Oscillation of sin wave is associated with its :**
   a) Cycle
   b) Amplitude
   c) Phase
3) **Which of the following is independent of the medium :**
   a) Frequency
   b) Wavelength
   c) Speed
4) **In a human voice :**
   a) The number of frequencies is infinite , but the range is limted
   b) The number of frequencies is finite , but the range is limted
   c) The number of frequencies is infinite , but the range is umlimted
5) **Bit length is measured by :**
   a) Bits
   b) Bits/second
   c) Meters

6) The main difference between periodic abd non-periodic digital signals is :

a) Bandwidth
b) Frequencies
c) Amplitude

7) A block of 256 sequential 12-bit data word is transmitted serially in 0.016s, Calculate the time duration of 1 word :

a) 1.33 μs
b) 62.5 μs
c) 16 μs

8) A PCM transmission requires :

a) More bandiwdth than AM
b) Less bandiwdth the AM
c) Same bandiwdth as AM

9) A bandwidth of :

a) 10 KHZ for transmission of speech signals
b) 200 KHZ for transmission of speech signals
c) 4 KHZ for transmission of speech signals

10) UHF propagation takes place primarily through :

a) Ground wave
b) Sky wave
c) Line-Of-Sight wave

11) Indicate which of the following system is digital :

a) PPM
b) PCM
c) PWM

12) A signal of 140 MHZ has a bandwidth of s a bandwidth ±20 MHZ, what is the Nyquist sampling rate :

a) 80 MHZ
b) 160 MHZ
c) 40 MHZ
d) 320 MHZ

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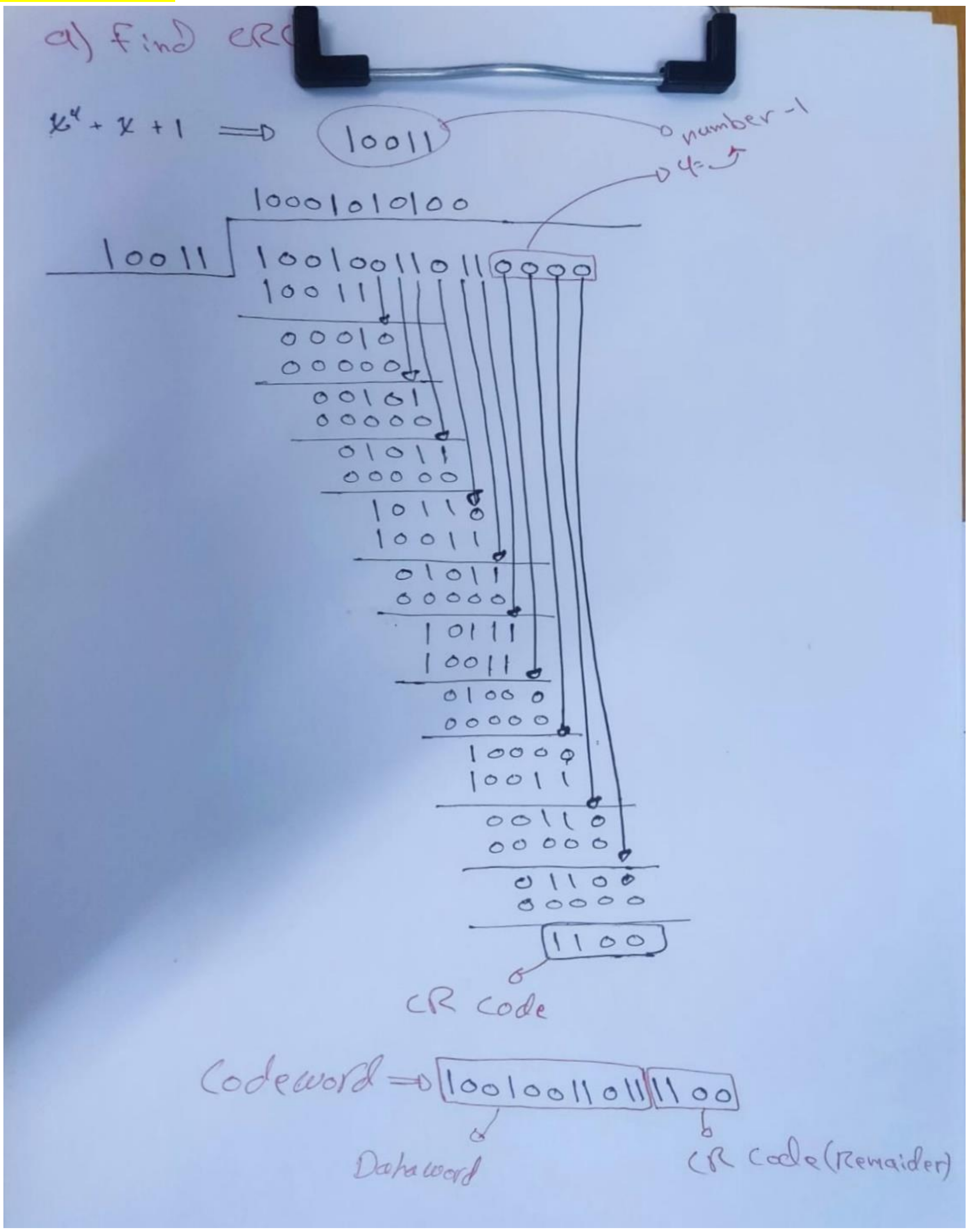
Q3 (10 marks) : If the data combination is represented as 10010011011 , Which will be augmented according to the generator ($x^4 + x + 1$)

> ***a) Find CRC .***
> ***b) If the received combination is ($x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^3 + x^2$) , what is the decoding result for this combination .***

SOLUTION :

a)



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b)

AT RECEIVER (DECODING) :

```
         11001110
       ___________
10011 | 110110111100
        10011
        -----
         10000
         10011
         -----
          00111
          00000
          -----
           01111
           00000
           -----
            11111
            10011
            -----
             11001
             10011
             -----
              10100
              10011
              -----
               01110
               00000
               -----
                1110
```

NOT ZERO , NOT ACCEPT

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Q4 (11 marks) : For 11 binary bit string 01001100011 :

a) ***Find the number of transistions for the encoding schemes Manchester & Diffrential Manchester .***

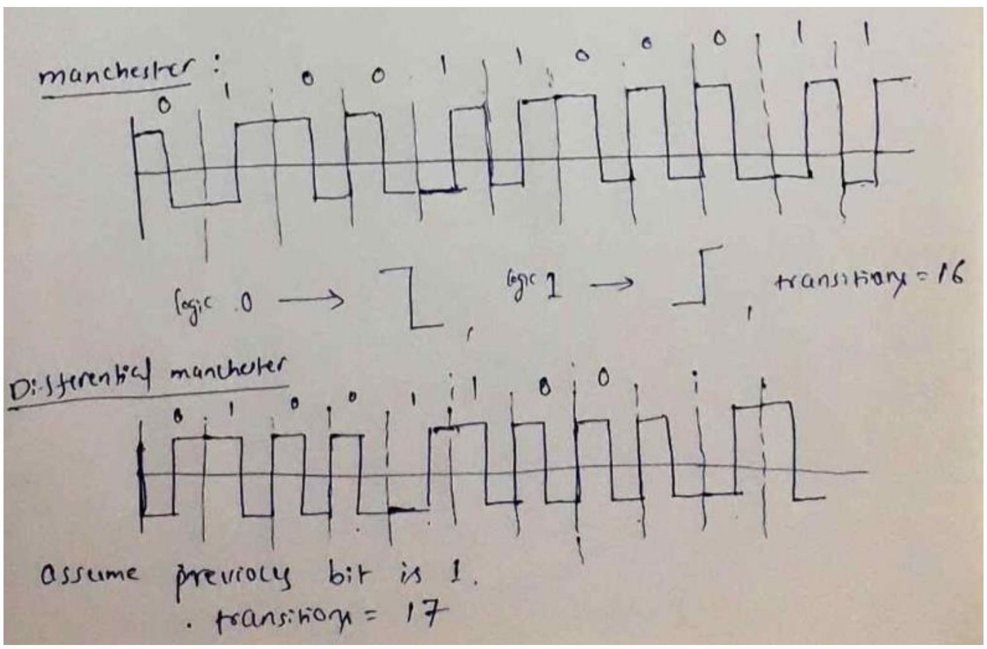
b) ***If the above combination rate is 10Mbps, Calculate the average signal rate using Manchester code .***

c) ***Calculate the number of data elements carried by each signal element in Diffrential Manchester code .***

d) ***Calculate maximum bandwidth .***

SOLUTION :

a-



b- (S = 10M baud rate)

c- (r = ½)

d- (Bmin = 10MHZ)

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- ***More Question On Data Communication & Networking***

Q1 : $F(t)=(12\cos(4t))^2$ ***, using identity :-***
$\cos x * \cos y = 0.5*(\cos(x-y) + \cos(x+y))$

a) ***Find Period .***
b) ***Find fs .***

a- $144*0.5(\cos(4t-4t) + \cos(4t+4t))$

$72(1 + \cos(8t))$

$72 + 72\cos(8t)$

$T = 2\pi/8$

a) $F_{max} = 8/2\pi$

$F_s = 2 * F_{max} = 8/\pi$

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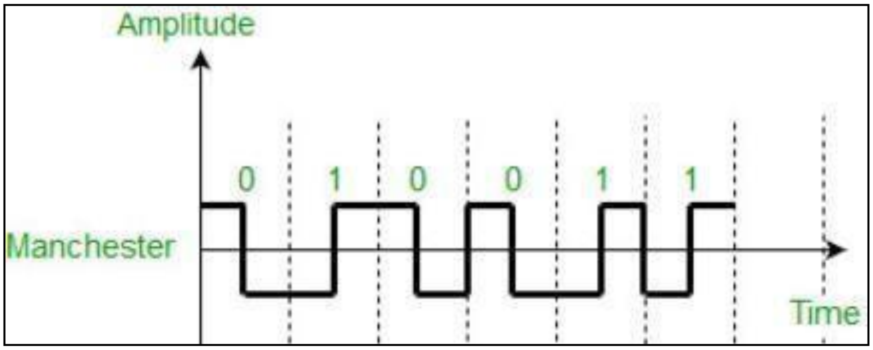
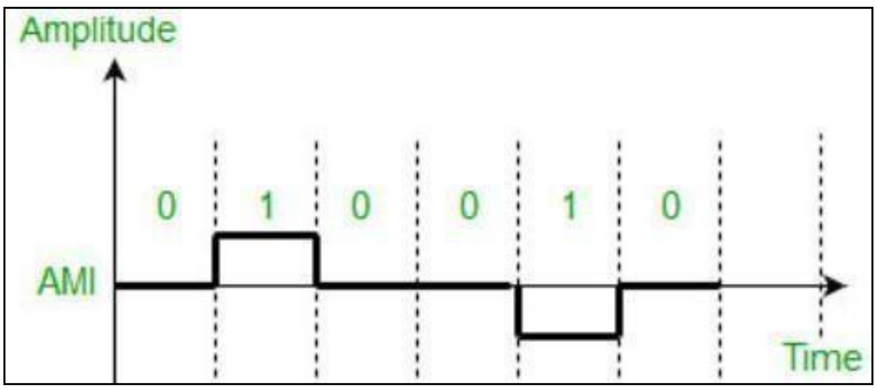
- ***More Question On Data Communication & Networking***

Q2 : For this two figure shown below :





a) ***Find bit per signal for each figure .***

((½) Fig.1 , (1) Fig.2)

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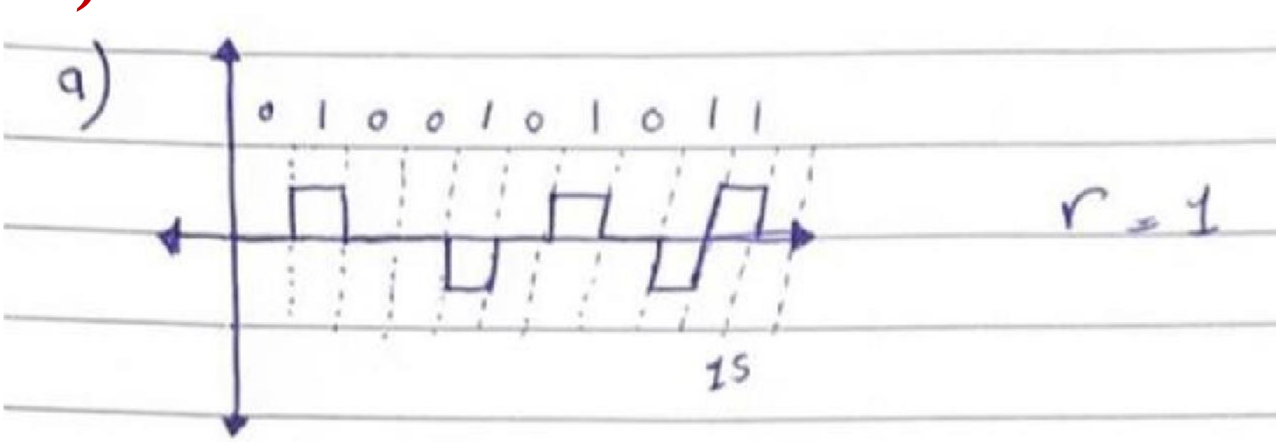
- *More Question On Data Communication & Networking*

Q3 : Consider the following sequence : (0100101011)

a) ***Represent the above sequence into bipolar-AMI waveform .***
b) ***Calculate the bit rate if the sequence is sent in one second .***
c) ***Calculate the baud rate of the transmitted sequence .***
d) ***Sketch the above bit stream into Manchester-Code .***
e) ***With respect to above results calculate minimum required bandwidth .***
f) ***Consider SNR=24 DB , Calculate the maximum channel capacity & Number of level FOR AMI .***

a)



b) ***N = 10 bps***

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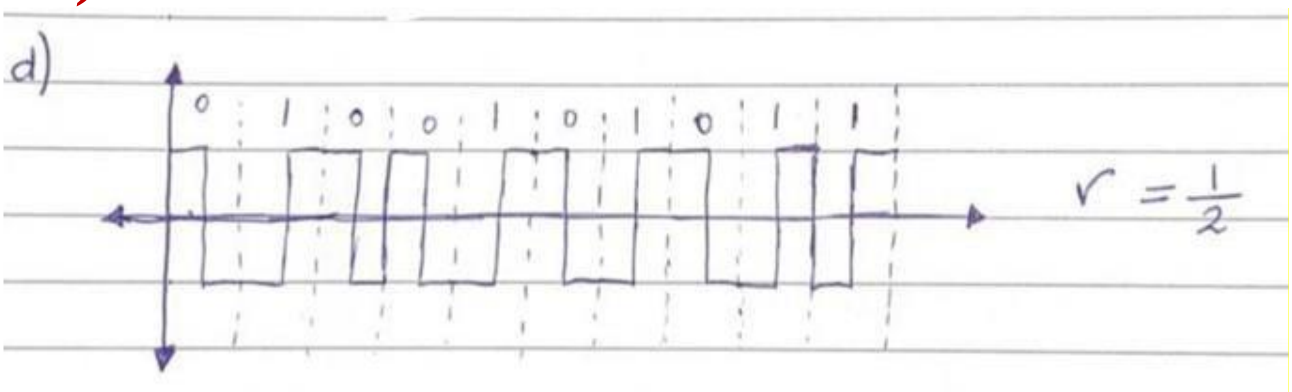
c) $S = N = 10$ ***baud rate***

d)



e) $B(min)\ [AMI] = S(avg) = \frac{1}{2} * N * 1/r$

$= \frac{1}{2} * 10 * 1/1 = 5\ HZ$

$B(min)\ [Manchester] = S(avg) = \frac{1}{2} * N * 1/r$

$= \frac{1}{2} * 10 * 1/\frac{1}{2} = 10\ HZ$

f) $SNR(db) = 24\ db$,, $SNR(db) = 10\ Log(\ SNR\ )$

$24 = 10\ Log\ (\ SNR\ ) \rightarrow SNR = 251.2$

$** \ C = B * Log_2(\ 1 + SNR\ )$

$C = 5 * Log_2(\ 1 + 251.2\ ) \rightarrow C = 39.89\ bps$

$** \ NyquistRate = 2 * B * Log_2(\ L\ )$

$39 = 2 * 5 * Log_2(\ L\ ) \rightarrow L = 14.9$

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